

205 9/28:

Nat , a set with "0" and a successor function $s(x): \text{Nat} \rightarrow \text{Nat}$ satisfying several rules.

Arithmetic ops:

$+$, $\cdot$ using "0" and " $s(x)$ ".

satisfies
• commutativity
• distributivity

Order structure:

Defⁿ: let F be a set, an order on F is a relation satisfying

(Friedrichs) 01 > For any $a, b \in F$, one of the following holds
 $a < b$, $a = b$, $b < a$

(Transitive) 02 > For any $a, b, c \in F$
 $a < b$ & $b < c$
 ~~$a < b$~~ $\Rightarrow a < c$

In general, a set F with an order " $<$ " is called an ordered set.

Nat: we can define an order in Nat . $\forall a, b \in \text{Nat}$
we say $a < b$ if $\exists c \in \text{Nat} \mid c \neq 0, a + c = b$

Summarize:
 $(\underbrace{\text{Nat}}_{\text{set}}, \underbrace{+, \cdot}_{\text{Arithmetic}}, \underbrace{<}_{\text{order}})$

Standard notation:

$$\text{Nat} = \{0, s(0), s(s(0)), \dots\} \cong \mathbb{N} = \{0, 1, 2, \dots\}$$
$$\Rightarrow (\mathbb{N}, +, \cdot, <)$$

Q: Given any $a, b \in \mathbb{N}$. can we always find $x \in \mathbb{N}$
 s.t. $x+a=b$
No!
 $a=0, b=1$

$$x+b = x+1 = x+1 = x+s(0) = s(x+0) = s(x)$$

$$x+1=0 \iff s(x)=0? \nexists x.$$

Enlarge $\mathbb{N}$:

when we enlarge, we keep:

- $\mathbb{N}$ is a subset
- keep the good structure

Note: x is determined by the "order pair" of natural #s (a, b) .

so we introduce

$$\mathbb{N} \times \mathbb{N} = \left\{ (a, b) : a, b \in \mathbb{N} \right\}$$

no algebraic structure yet

OBSERVE:

different order pair may give same x .

$$(a, b) = (2, 1) \quad \text{and} \quad (3, 2)$$

$$\begin{aligned} x+1 &= 2 \\ x+s(0) & \\ s(x+0) & \\ s(x) &= 2 \Rightarrow x=1 \end{aligned}$$

$$\begin{aligned} x+2 &= 3 \\ x+s(1) & \\ s(x+1) & \\ s(x+s(0)) & \\ s(s(x+0)) & \\ s(s(x)) &= 3 \Rightarrow x=1 \end{aligned}$$

Next, we define an "equivalence relation" on $\mathbb{N} \times \mathbb{N}$

$$(m, n) \sim (p, q) \iff m+q = n+p$$

addition in $\mathbb{N}$

check: $\sim$ is equivalence relation

(I) Reflexive

$$(m, n) \sim (m, n)$$

(II) symmetric

$$(m, n) \sim (p, q) \Rightarrow (p, q) \sim (m, n)$$

(III) Transitive:

$$(m, n) \sim (p, q) \text{ \& } (p, q) \sim (r, s) \Rightarrow (m, n) \sim (r, s)$$

(ϵ_0 !)

Define:

The integer set

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim = \left\{ [(m, n)] \mid m, n \in \mathbb{N} \right\}$$

(1) $\mathbb{N} \hookrightarrow \mathbb{Z}$

$\forall n \in \mathbb{N}$

$$n \longleftrightarrow [(n, 0)]$$

so we identify $\mathbb{N}$

$\uparrow$
 $\sim$
 $\uparrow$
 $\mathbb{N}$

$$x + 0 = x$$

as the image of the above map, which is a subset of $\mathbb{Z}$

• Arith. ops:

• addition $+_{\mathbb{Z}}$:

$$[(m, n)] +_{\mathbb{Z}} [(p, q)] := [(m+p, n+q)]$$

• Multiplication $\cdot_{\mathbb{Z}}$:

$$[(m, n)] \cdot_{\mathbb{Z}} [(p, q)] := [(m \cdot p + n \cdot q, m \cdot q + n \cdot p)]$$

From $\mathbb{N}$

Well defined:

eg. for $+\mathbb{Z}$: we have to make sure if

$$(m, n) \sim (m', n') \quad \& \quad (p, q) \sim (p', q')$$

we have

$$[(m, n)] + [(p, q)] = [(m', n')] + [(p', q')]$$

(Ex!)

$$(\mathbb{Z}, +_{\mathbb{Z}}, \cdot_{\mathbb{Z}}) \Big|_{\mathbb{N}} \cong (\mathbb{N}, +, \cdot)$$

For addition:

$$m, p \in \mathbb{N}$$

$$\underline{m+p}: \quad m \longleftrightarrow [(m, 0)] \in \mathbb{Z}$$

$$p \longleftrightarrow [(p, 0)] \in \mathbb{Z}$$

$$[(m, 0)] +_{\mathbb{Z}} [(p, 0)] = [(m+p, 0)] \longleftrightarrow m+p$$

$$\text{So, } (\mathbb{N}, +, \cdot) \hookrightarrow (\mathbb{Z}, +_{\mathbb{Z}}, \cdot_{\mathbb{Z}})$$

$$n \longrightarrow [(n, 0)]$$

Recall: why do we want to enlarge $\mathbb{N}$?

WTF ~ solution to $\boxed{x+b=a}$ $a, b \in \mathbb{N}$

check: $\mathbb{Z}$ is the proper set to include all the solutions

Take $[(m, n)], [(p, q)] \in \mathbb{Z}$

define: $x = [(m+q, n+p)] \in \mathbb{Z}$

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check

$$x + [(p, q)] = [(m, n)]$$

"

//? ✓

$$[(m+q+p, n+p+q)]$$

$$(m, n) \sim (r, s)$$

$$\Rightarrow m+s = n+r$$

This allows us to introduce a "new" operation in $\mathbb{Z}$
subtraction:

$$[(m, n)] - [(p, q)] := [(m+q, n+p)]$$

Using this "new" operation, we can identify

$$-n \longleftrightarrow [(0, n)]$$

"identification"

$$n \longleftrightarrow [(n, 0)]$$

"Traditional Notation of $\mathbb{Z}$ "

$$\mathbb{Z} = \{ [(m, n)] : m, n \in \mathbb{N} \}$$

$$[(m, n)] = \begin{cases} [(m-n, 0)] & \text{if } m \geq n \\ [(0, n-m)] & \text{if } n \geq m \end{cases}$$

$$[(1, 1)] + [(1, 2)] = [(1, 0)]$$

$$\mathbb{Z} \longleftrightarrow \{ 0, \pm 1, \pm 2, \dots \}$$

ordered structure on $\mathbb{Z}$:

$\forall [m, n], [p, q] \in \mathbb{Z}$. we define

$$[m, n] < [p, q] \text{ if } m+q < n+p$$

check: This is an "order"

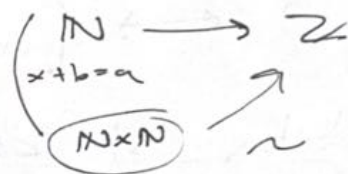
Summary: $(\mathbb{Z}, +, \cdot, <) \supseteq (\mathbb{N}, +, \cdot, <)$

Q: For any $a, b \in \mathbb{Z}$, with $a \neq 0$. can we find $x \in \mathbb{Z} \mid a \cdot x + b = 0$?

In general, No.

Next, we need to enlarge $\mathbb{Z}$.

Introduce a set:



$$\mathbb{Q} := \left\{ (b, a) : b, a \in \mathbb{Z}, a \neq 0 \right\}$$

Goal: observe an equivalence relation

Def: we say (b, a) & $(m, n) \in \mathbb{Q}$ are equivalent if

$$(b, a) \sim (m, n) \iff b \cdot_2 n = a \cdot_2 m$$

Def: The rational number set

$$\mathbb{Q} := \mathbb{Q} / \sim = \left\{ [(b, a)] : b, a \in \mathbb{Z}, a \neq 0 \right\}$$

• $\mathbb{Z} \subset \mathbb{Q}$.

we can identify $\frac{m}{a} \in \mathbb{Q}$ with $[\frac{m}{a}, 1]$.

$$\frac{m}{a} \mapsto [\frac{m}{a}, 1]$$

$\downarrow \quad \uparrow$
 $b \quad a$

$$a \cdot x = b \rightsquigarrow 1 \cdot x = m$$

$$\mathbb{N} \subset \mathbb{Z}$$

$$n \mapsto [n, 0]$$

$$x + 0 = n$$

• Ans ops :

→ want new operation to recover op from $\mathbb{Z}$

$$+_Q : [(b, a)] +_Q [(m, n)] := [(b \cdot n + a \cdot m, a \cdot n)]$$

$$\cdot_Q : [(b, a)] \cdot_Q [(m, n)] := [(b \cdot m, a \cdot n)]$$

• check: $+_Q, \cdot_Q$ recovers $+, \cdot$ in $\mathbb{Z}$ when restricted to $\mathbb{Z}$.

something "new":

proposition: For any $[(x, y)] \in \mathbb{Q}$

$$\bullet [(x, y)] + [(0, 1)] = [(x, y)]$$

intro/def $\bullet [(x, y)] \cdot [(1, 1)] = [(x, y)]$

$$\bullet [(x, y)] \cdot [(y, x)] = [(1, 1)] \text{ if } x, y \neq 0.$$

implies:

(I) $[(0, 1)]$ is the identity for addition

(II) $[(1, 1)]$ is the identity for $\cdot$

(III) $[(x, y)] \in \mathbb{Q}$ has an "inverse" for multiplication.

Theorem: $(\mathbb{Q}, +, \cdot)$ is a field!